GC-biased gene conversion, the recombination requirement, and Lewontin's paradox of variation

Erik Garrison

Draft. Lean 4 formalization: github.com/ekg/lewontin-paradox

Abstract. Lewontin's paradox — that neutral nucleotide diversity π varies by only 2 orders of magnitude across metazoans whose census population sizes span 7 — is unresolved; current linked-selection models do not fully flatten the diversity–size relationship (Hermisson & Pfanner 2024). We argue that the standard mutation–selection–drift balance at W/S sites under GC-biased gene conversion (gBGC) supplies the missing flattening: in large populations it saturates to an N_e-independent floor $2\mu/(u\delta)$, where u is the per-site conversion-coverage rate and δ the per-conversion GC bias. The bias is always present ($\delta > 0$, inherent to mismatch repair) and is maintained by selection for the sequence homology that meiotic recombination requires. The resulting floor is the form previously (and incorrectly) reached via "gene-conversion homogenization" arguments; here it is derived by the correct mutation–selection–drift route. The argument — including the marginal-coalescent refutation of the unbiased case and the self-limiting saturation — is formalized and machine-checked in a Lean 4 library.

1. Introduction

Neutral theory predicts equilibrium diversity $\pi = 4N_e\mu$ at a neutral site, scaling linearly with effective population size N_e. Yet across metazoans, while census sizes N span roughly 10^2 to 10^{10}, within-species π ranges only 10^{-4} to 10^{-2} — the "paradox of variation" first noted by Lewontin (1974). The leading candidate resolution, selection at linked sites (hitchhiking and background selection), has recently been shown quantitatively insufficient to flatten the relationship (Hermisson & Pfanner 2024); weak genetic draft from sparse sweeps gives a sublinear but not flat scaling (Achaz & Schertzer 2023).

A recurring but under-developed alternative appeals to non-crossover gene conversion. The intuition: gene conversion is a **copying** mechanism (it copies a donor sequence onto a recipient), and copies between identical-by-descent (IBD) sequences are molecularly silent, so the observable conversion rate scales with standing heterozygosity — a "self-regulating homogenization" that, it is claimed, drives an equilibrium $\pi = \mu/(c \cdot L)$ decoupled from N_e (with c the conversion-initiation rate, L the tract length). We show here that this specific argument is mathematically unsound, but that the destination it seeks — an N_e-independent diversity floor in large populations — is nonetheless **correct**, reached instead by the standard mutation–selection–drift balance at W/S sites under gBGC, with the recombination requirement supplying the evolutionary reason the relevant parameters are maintained.

2. Unbiased gene conversion is a martingale: the marginal coalescent

Consider a biallelic `Aa` heterozygote and a gene-conversion tract spanning the focal site. With no DSB asymmetry, the DSB falls on each homolog with probability 1/2; the donor is the uncut homolog. A **single gamete** from this meiosis carries `A` with probability

$$\mathrm{P}(\mathrm{A} \mid \mathrm{Aa}, \text{conversion}) = \frac{1}{2} \cdot \frac{1}{4} + \frac{1}{2} \cdot \frac{3}{4} = \frac{1}{2},$$

exactly Mendelian. The per-gamete variance is therefore 1/4 with or without conversion, the binomial sampling variance $2Np(1-p)$ is unchanged, and conversion contributes **zero** drift variance at a single site. This is the forward-time face of the coalescent result (Wiuf & Hein 2000): in the coalescent with gene conversion, the **marginal** genealogy at any single site is the Kingman coalescent, rate $1/(2N_e)$, independent of the conversion rate c. Gene conversion reshuffles the **correlations between sites** (linkage disequilibrium, the ancestral recombination graph); it does not change the **marginal** diversity. Hence

$$\mathrm{E}[\pi] = 2\mu \cdot \mathrm{E}[T_{\mathrm{MRCA}}] = 2\mu \cdot 2N_e = 4N_e\mu, \quad c-\text{independent}.$$

The asserted diversity drain $c \cdot L \cdot H^2$ of the "homogenization" argument is therefore not a valid population-genetic loss term: a conversion that copies a homozygote across a heterozygous site changes the **gamete**, not the **population** allele frequency. (Lean: `Transmission.lean`, `Coalescent.lean`.)

3. GC-biased gene conversion is always on and selection-like

The unbiased case $\delta = 0$ (where $\delta = b - \frac{1}{2}$, $b = \mathrm{P}(\text{transmit GC} \mid \mathrm{W}\backslash/\mathrm{S}, \text{conversion})$) is a theoretical limit, not the real regime. The GC bias is inherent to mismatch repair of heteroduplex mismatches; empirically $b \approx 0.68$ ($\delta \approx 0.18$, often higher in hotspots). Any $\delta > 0$ breaks the martingale and makes gBGC a directional force favoring GC, with a **discontinuity at zero**: the scaled strength is zero iff $\delta = 0$, and positive otherwise.

Because gBGC only manifests when a conversion tract spans the site, the effective per-site per-generation selection coefficient is

$$s_{\text{eff}} = u \cdot \delta, \quad \gamma = 4N_e u\delta,$$

where $u = c \cdot L$ is the per-site conversion-coverage rate (the probability a site lies in a conversion tract per generation). This is **linear in** N_e — selection-like — so gBGC compresses the large-N_e end of the diversity–size relationship (the right direction for the paradox), in contrast to classic background selection, which for strong deleterious selection is **N_e-independent** and thus does not compress the **scaling** (Corbett-Detig et al. 2015). (Lean: `GBGC.lean`.)

A naive estimate with the **genome-wide** $u \approx 1.8 \times 10^{-3}$ and $\delta \approx 0.18$ gives enormous γ in every species (e.g. $\gamma \approx 13$ in humans, $\approx 1.3 \times 10^3$ in *Drosophila*), which would crush π far below observation. That this does not occur is the self-limitation of gBGC: its rate is proportional to the W/S heterozygosity $2f(1-f)$ and vanishes at fixation — gBGC "wins" by depleting AT, at which point it has no substrate and stops. (Lean: `Saturation.lean`.)

4. The recombination requirement: why $\delta > 0$ and u are maintained

The flattening above requires $\delta > 0$ and substantial u; these are not accidents. Meiotic recombination (DSB repair) is essential for chromromosome segregation and fertility. Non-crossover resolution via synthesis-dependent strand annealing or a double-Holliday-junction intermediate requires the invading strand to form a stable heteroduplex and for the junction to **migrate** away from the DSB; branch migration requires near-identity, since mismatches impede it. Thus conversion only occurs between near-identical sequences, and conversion **makes** sequences more identical — a positive feedback (homology enables conversion, conversion maintains homology). Because recombination is essential, selection maintains the homology that conversion maintains, keeping the genome in a "recombino-genic" state and, with it, the GC-biased repair machinery ($\delta > 0$) and high conversion rates (u). This is the evolutionary cause of the parameters that make the saturation bite, not a separate diversity-reducing force. (Lean: `RecombinationHomology.lean`.)

5. Mutation–selection–drift balance at W/S sites: the saturation

With $\delta > 0$ always, gBGC acts as genic selection **against** the AT allele at every W/S site, with $s = u\delta$. In a finite population this is the textbook mutation–selection–drift balance, with two regimes:

$$\text{weak gBGC } (2N_e u\delta \ll 1): \quad \pi_{\mathrm{W\backslash/S}} \approx 4N_e\mu \quad (\text{neutral, } N_e \text{-scaled});$$

$$\text{strong gBGC } (2N_e u\delta \gg 1): \quad \pi_{\mathrm{W\backslash/S}} \approx 2\mu/(u\delta) \quad (N_e \text{-independent, saturated}).$$

The crossover is at $N_e \approx 1/(2u\delta)$. In the strong regime the AT allele is held at the mutation–selection balance frequency $q^* = \mu/s = \mu/(u\delta)$, so the heterozygosity $2q^*(1-q^*) \approx 2\mu/(u\delta)$. This floor is (up to a factor of 2 and the bias δ) the form $\mu/(c \cdot L)$ sought by the homogenization argument — but it is reached by mutation–selection–drift balance at W/S sites, **not** by the flawed "IBD homogenization" drain. The saturation is N_e-independent and is always on (since $\delta > 0$); for N_e past the crossover it pins W/S diversity at the floor, below the neutral expectation. (Lean: `MutationSelectionDrift.lean`, `Composition.lean`.)

6. Magnitude and the effective conversion rate

For the floor to address the paradox, it must lie at the observed diversity magnitude in large-N_e species. With an effective per-site coverage $u_{\mathrm{eff}} \approx 10^{-5}$ (a typical, non-hotspot rate — note the genome-wide average $\approx 1.8 \times 10^{-3}$ is dominated by hotspots), $\mu \approx 10^{-8}$ and $\delta \approx 0.18$,

$$2\mu/(u_{\mathrm{eff}}\delta) \approx 2 \times 10^{-8}/(10^{-5} \times 0.18) \approx 0.01,$$

matching observed π in large-N_e species. The floor is sensitive to u_{eff}, which is precisely the quantity left uncertain by the "silent-conversion" measurement problem: most conversion tracts fall in IBD sequence and leave no molecular footprint, so published genome-wide rates are lower bounds on the true rate and overstate the average effect. Quantifying u_{eff} across the genome (from, e.g., UK Biobank local-heterozygosity versus detected-tract-rate regressions) is the key empirical input. (Lean: `ObservableFraction.lean`, `Bounds.lean`.)

7. Limits and open questions

The saturation flattens **W/S-site** diversity; genome-wide flattening is partial:

- **W/S fraction.** Only sites that are (or recently were) W/S polymorphisms feel gBGC directly. A partial W/S fraction with an N_e-independent floor and a neutral remainder leaves a mix $\pi = (1 - X) \cdot 4N_e\mu + X \cdot (2\mu/(u\delta))$, which is flat only if $X = 1$ (Lean: `GenomeMix.lean`). The linked neutral reduction from gBGC at W/S sites broadens the effect.
- **Effective u.** The floor magnitude hinges on u_{eff}, uncertain by orders of magnitude and concentrated in hotspots.
- **Saturation vs. over-compression.** With the genome-wide u, γ is huge; the self-limitation and hotspot concentration are what keep the bite finite. The distribution of local u sets the genome-wide average.
- **Comparables.** Whether gBGC's N_e-scaled saturation can be distinguished from, or combined with, weak genetic draft (Achaz & Schertzer 2023) and reassessed linked selection (Hermisson & Pfanner 2024) is an open empirical question.

8. The formalization

The argument is machine-checked in a 16-module Lean 4 library (depending on Mathlib) at `lean/src/LewontinParadox`. It contains, with no `sorry`, the marginal-coalescent invariance and the master refutation of the unbiased "homogenization" claim, the gBGC strength and its N_e -linearity, the composition equilibrium, the self-limiting saturation, the mutation–selection–drift crossover, the linkage-based distinction between repeat and single-copy diversity, the genome-mix flatness theorem, and the recombination-homology feedback. The single imported hypothesis is the Kingman mean $E[T_{\mathrm{MRCA}}] = 2N_e$; everything else is proven. The build is reproducible:

```
cd lean && lake build
```

9. References

- Achaz G, Schertzer E (2023). Weak genetic draft and Lewontin's paradox. *bioRxiv* 2023.07.19.549703.
- Corbett-Detig R et al. (2015). Natural selection constrains neutral diversity across a wide range of species. *PLOS Biol* 13: e1002112.
- Hermisson J, Pfanner C (2024). Quantifying the relationship between genetic diversity and population size… *eLife* 67509.
- Lewontin RC (1974). *The Genetic Basis of Evolutionary Change.* Columbia Univ. Press.
- Wiuf C, Hein J (2000). The coalescent with gene conversion. *Genetics* 155: 451–462.
- Cole F, Jasin M, Keeney S (2012). Preaching about the converted. *DNA Repair* 11: 617.